Practice Set 4 — Applied

Name: _____ Student ID: _____

Part A: 10 × 2 points · Part B: 10 × 3 points · Part C: 10 × 5 points · Total: 100 points

Useful: Φ(1)=0.8413, Φ(1.5)=0.9332, Φ(2)=0.9772, z0.90=1.2816, z0.95=1.645, e^(-2)=0.1353, e^(-1.2)=0.3010. Quartile position Lp=(p/100)(n+1).

Part A · Multiple Choice (20 points)

MC1. Among 200 farms, 60 received a subsidy. If subsidy status is coded 1/0, what is its sample mean?

(a) 0.20
(b) 0.30
(c) 0.60
(d) 1.00

MC2. The first and third quartiles are Q1=14 and Q3=22. What is the upper 1.5×IQR fence?

(a) 26
(b) 30
(c) 34
(d) 38

MC3. Independent events A and B have P(A)=0.40 and P(B)=0.25. Find P(A ∪ B).

(a) 0.10
(b) 0.55
(c) 0.65
(d) 0.75

MC4. P(disease)=0.20, P(+ | disease)=0.90, P(+ | healthy)=0.10. Find P(disease | +).

(a) 0.1800
(b) 0.6000
(c) 0.6923
(d) 0.9000

MC5. X ~ Bernoulli(0.30). What is Var(X)?

(a) 0.09
(b) 0.21
(c) 0.30
(d) 0.70

MC6. X ~ Binomial(5,0.4). Find P(X=2).

(a) 0.2304
(b) 0.2592
(c) 0.3456
(d) 0.4000

MC7. Hourly truck arrivals follow Poisson(2). Approximate P(no trucks in one hour).

(a) 0.1353
(b) 0.2707
(c) 0.3679
(d) 0.8647

MC8. Delivery time T ~ Uniform(10,30). Find P(T > 25).

(a) 0.10
(b) 0.25
(c) 0.50
(d) 0.75

MC9. T ~ Normal(50,5²). Find P(T < 55), given Φ(1)=0.8413.

(a) 0.1587
(b) 0.5000
(c) 0.8413
(d) 0.9772

MC10. X has sd(X)=6, and Y=3X 4. Find sd(Y).

(a) 6
(b) 14
(c) 18
(d) 36

Part B · True / False (30 points)

TF1. The mean of a binary 0/1 variable equals the proportion coded 1.

☐ True ☐ False

Explanation: _____

TF2. For a valid PMF, the probabilities can sum to more than one if there are many outcomes.

☐ True ☐ False

Explanation: _____

TF3. The variance of Bernoulli(p) is largest at p=0.5.

☐ True ☐ False

Explanation: _____

TF4. A binomial model is valid even if success probabilities differ from trial to trial.

☐ True ☐ False

Explanation: _____

TF5. If events A and B are independent, $P(A \cap B)=P(A)P(B)$.

☐ True ☐ False

Explanation: _____

TF6. If $X \sim \text{Poisson}(\lambda)$, then $E(X)=\lambda$ but $\text{Var}(X)=\lambda^2$.

☐ True ☐ False

Explanation: _____

TF7. Adding a constant to a random variable changes its expected value but not its variance.

☐ True ☐ False

Explanation: _____

TF8. $P(A \mid B)$ and $P(B \mid A)$ always have equal values.

☐ True ☐ False

Explanation: _____

TF9. For a continuous random variable, the probability of one exact point is zero.

☐ True ☐ False

Explanation: _____

TF10. A positive correlation between two variables proves that one causes the other.

☐ True ☐ False

Explanation: _____

Part C · Short Answers (50 points)

C1. Daily visitors to a produce market (sorted): 12, 14, 15, 17, 19, 21, 23, 39.

(a) [5 pt] Compute the sample mean and median. What does their comparison suggest?

Work: _____

_____

(b) [5 pt] Find Q1 and Q3 using Lp=(p/100)(n+1) with linear interpolation. Apply the upper 1.5×IQR fence.

Work: _____

_____

(c) [5 pt] Calculate sample variance, sample sd, and coefficient of variation.

Work: _____

_____

C2. Three processing lines supply 50%, 30%, and 20% of goods. Their defect rates are 2%, 5%, and 10%, respectively.

(a) [5 pt] Find the unconditional probability that an item is defective.

Work: _____

_____

(b) [5 pt] Given that a randomly sampled item is defective, find P(Line 3 | defective).

Work: _____

_____

(c) [5 pt] Given that an item is not defective, find P(Line 1 | not defective).

Work: _____

_____

C3. A farmer receives X orders per day. The PMF is x=0,1,2,3 with probabilities 0.1,0.2,0.5,0.2.

(a) [5 pt] Find E(X), $E(X^2)$, Var(X), and sd(X).

Work: _____

_____

(b) [5 pt] Profit is Y=25X 8 (thousand won). Find E(Y) and sd(Y).

Work: _____

_____

(c) [5 pt] Four customers independently pay late with probability .20 each. Find P(at least one late payment).

Work: _____

_____

(d) [5 pt] Waiting time T follows an exponential distribution with rate .1 per minute. Find P(T>20) and P(T>30 | T>10).

Work: _____

_____

ANSWER KEY & STEP-BY-STEP SOLUTIONS

Part A · Multiple Choice

MC1 · B

더미변수의 평균은 1인 비율: 60/200=0.30.

MC2 · C

IQR=22 14=8. 상한=22+1.5×8=34.

MC3 · B

독립이므로 P(A∩B)=0.4×0.25=0.10. 합집합=.4+.25 .10=.55.

MC4 · C

베이즈: .20×.90/[.20×.90+.80×.10]=.18/.26≈.6923.

MC5 · B

Var(X)=p(1 p)=.3×.7=.21.

MC6 · C

C(5,2)×.4²×.6³=10×.16×.216=.3456.

MC7 · A

P(X=0)=e^(2)=.1353.

MC8 · B

(30 25)/(30 10)=5/20=.25.

MC9 · C

z=(55 50)/5=1 → Φ(1)=.8413.

MC10 · C

sd(3X 4)=|3|sd(X)=18.

Part B · True / False

TF1 · TRUE

합계 ΣX 는 1의 개수이므로 평균 ΣX /n은 1의 비율.

TF2 · FALSE

PMF의 가능한 모든 값에 대한 확률합은 반드시 정확히 1.

TF3 · TRUE

p(1 p)=0.25 (p 0.5)²이므로 최대는 0.25.

TF4 · FALSE

독립 시행들의 성공확률 p가 모두 같아야 통상적 Binomial(n,p) 모델.

TF5 · TRUE

독립성의 곱셈법칙.

TF6 · FALSE

포아송은 기댓값과 분산 모두 λ.

TF7 · TRUE

$E(X+c)=E(X)+c$, $Var(X+c)=Var(X)$.

TF8 · FALSE

조건부확률의 분모가 서로 다르며 베이즈 정리로 연결.

TF9 · TRUE

연속형 확률은 구간에 대한 밀도 적분으로 계산.

TF10 · FALSE

교란변수·역인과성 때문에 상관만으로 인과관계를 결론 내릴 수 없다.

Part C · Worked Solutions

C1 · Daily visitors to a produce market (sorted): 12, 14, 15, 17, 19, 21, 23, 39.

(a) Compute the sample mean and median. What does their comparison suggest?

합계=160, n=8 → 평균=20. 중앙값=(17+19)/2=18. 큰 값 39가 평균을 끌어올려 우측 치우침을 시사.

(b) Find Q1 and Q3 using $L_p=(p/100)(n+1)$ with linear interpolation. Apply the upper 1.5×IQR fence.

$L_{25}=2.25$ → $Q1=14+0.25(15\ 14)=14.25$. $L_{75}=6.75$ → $Q3=21+0.75(23\ 21)=22.5$. IQR=8.25. 상한 $=22.5+1.5(8.25)=34.875$; 39는 잠재적 이상치.

(c) Calculate sample variance, sample sd, and coefficient of variation.

평균 20. 제곱편차합=64+36+25+9+1+1+9+361=506. $s^2=506/7\approx72.2857$, $s\approx8.5021$, $CV\approx8.5021/20\times100\approx42.51\%$.

C2 · Three processing lines supply 50%, 30%, and 20% of goods. Their defect rates are 2%, 5%, and 10%, respectively.

(a) Find the unconditional probability that an item is defective.

$P(D)=.50(.02)+.30(.05)+.20(.10)=.01+.015+.02=.045$ (4.5%).

(b) Given that a randomly sampled item is defective, find P(Line 3 | defective).

Bayes: $P(L3|D)=P(D|L3)P(L3)/P(D)=.10\times.20/.045=.02/.045=4/9\approx.4444$.

(c) Given that an item is not defective, find P(Line 1 | not defective).

$P(L1|D\ )=.50(.98)/P(D\ )=.49/.955\approx.5131$.

C3 · A farmer receives X orders per day. The PMF is x=0,1,2,3 with probabilities 0.1,0.2,0.5,0.2.

(a) Find E(X), $E(X^2)$, Var(X), and sd(X).

$E(X)=0+.2+1+.6=1.8$. $E(X^2)=.2+4(.5)+9(.2)=4.0$. $Var(X)=4\ 1.8^2=.76$. $sd\approx.8718$.

(b) Profit is Y=25X 8 (thousand won). Find E(Y) and sd(Y).

E(Y)=25(1.8) 8=37 (천 원). sd(Y)=25√.76≈21.79 (천 원).

(c) Four customers independently pay late with probability .20 each. Find P(at least one late payment).

K~Binomial(4,.2). P(K≥1)=1 P(K=0)=1 .8⁴=1 .4096=.5904.

(d) Waiting time T follows an exponential distribution with rate .1 per minute. Find P(T>20) and P(T>30 | T>10).

P(T>20)=e^(.1×20)=e^(2)≈.1353. 무기억성: P(T>30|T>10)=P(T>20)=e^(2)≈.1353.